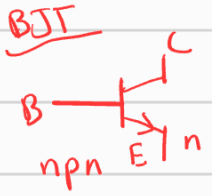
BJT



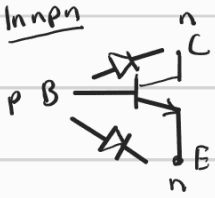
$$i_C = I_S e^{V_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} \qquad i_E = \frac{\beta+1}{\beta} i_C = \frac{1}{\alpha} i_C$$

$$i_E = (\beta+1) i_B$$

	Active	Sat	Cutoff
EBJ	Forward bias	Forward bias	Reverse bias
CBJ	Reverse bias	forward bias	Reverse bias

Innpn

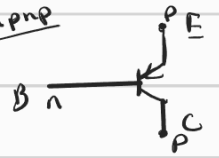


Active: $V_{BE} \geq 0,7$, $V_{BC} \leq 0,4V$

Cutoff: $V_{BE} < 0,7$, $V_{BC} \leq 0,4V$

Saturation: $V_{BE} \geq 0,7$, $V_{BC} > 0,4V$

Inpnp



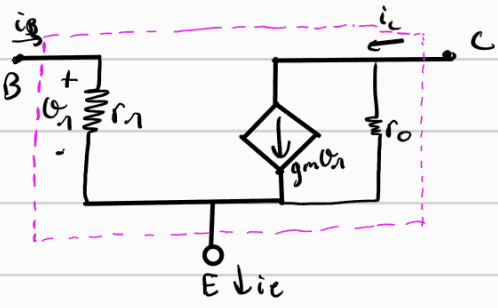
Active: $V_{EB} \geq 0,7$, $V_{CB} \leq 0,4V$

Cutoff: $V_{EB} < 0,7$, $V_{CB} \leq 0,4V$

Saturation: $V_{EB} \geq 0,7V$, $V_{CB} > 0,4V$

Small signal Models





$$i_b = \frac{v_\pi}{r_\pi}$$

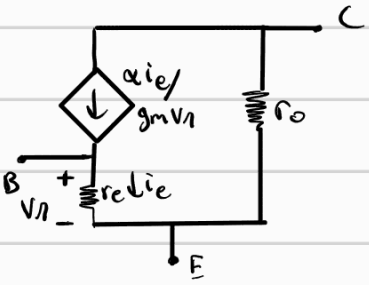
$r_o \rightarrow$ output resistance of bjt: $r_o = \frac{|V_A|}{I_C}$ → ac

$g_m \rightarrow$ transcunductance: $g_m = \frac{I_C}{V_T}$

$r_\pi \rightarrow$ base resistance: $r_\pi = (\beta+1) r_e$

$r_e \rightarrow$ emitter resistance: $r_e = \frac{V_T}{I_E} = \alpha \frac{V_T}{I_C} = \alpha \frac{1}{g_m}$

Tmodel

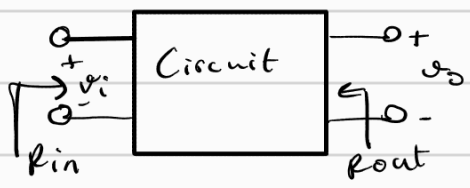


$$r_o = \frac{|V_A|}{I_C}$$

$$g_m = \frac{I_C}{V_T}$$

$$r_\pi = (\beta+1) r_e$$

$$r_e = \frac{V_T}{I_E} = \alpha \frac{V_T}{I_C} = \alpha \frac{1}{g_m}$$



Voltage Gain

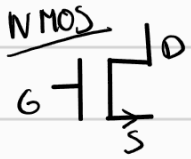
$$A_v = \frac{v_o}{v_i}$$

$R_{in} \rightarrow$ Input Resistance

$R_{out} \rightarrow$ Output Res.

Mosfet

NMOS



$V_{DS} < V_{GS} - V_t$ } Triode

$V_{GS} \geq V_t$ }

$V_{DS} \geq V_{GS} - V_t$ } Saturation

$V_{GS} < V_t$ } cutoff

$V_{OV} = V_{GS} - V_t$

NMOS

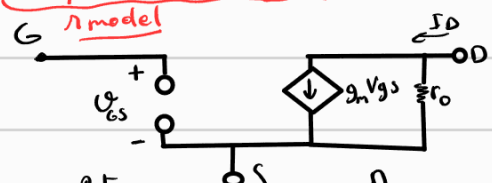
- Triode: $v_{GS} \geq V_t$
  - $V_{GD} > V_t$
  - $V_{DS} < V_{OV}$
  - $i_D = k'_n \left(\frac{W}{L}\right)\left(V_{OV} - \frac{1}{2}V_{DS}\right)V_{DS}$
- Saturation Region
  - $V_{GD} \leq V_t$
  - $V_{DS} \geq V_{OV}$
  - $i_D = \frac{1}{2} k'_n \left(\frac{W}{L}\right) v_{OV}^2$

$k'_n = \mu_n C_{ox}$

PMOS

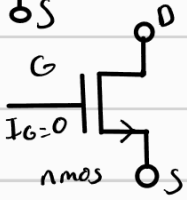
- Triode Region
  - $V_{DG} > |V_t|$
  - $V_{SD} < |V_{OV}|$
  - $i_D = k'_p \left(\frac{W}{L}\right)\left(|V_{OV}| - \frac{1}{2}V_{SD}\right)V_{SD}$
- Saturation Region
  - $V_{DG} \leq |V_t|$
  - $V_{SD} \geq |V_{OV}|$
  - $i_D = \frac{1}{2} k'_p \left(\frac{W}{L}\right) V_{OV}^2$

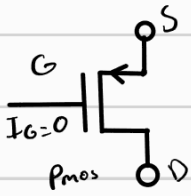
Mosfet small signal



$$g_m = \frac{2I_D}{V_{OV}}$$

$$r_o = \frac{V_A}{I_D}$$





Tmodel

